rearrange the complex conjugate elements of the matrix **M**. The notation for this rearrangement is $\mathbf{M}^\dagger$, as explained below. Our new equation is

$$\langle A|\mathbf{M}^\dagger = \begin{pmatrix} \alpha_1^* & \alpha_2^* & \alpha_3^* \end{pmatrix} \begin{pmatrix} m_{11}^* & m_{21}^* & m_{31}^* \\ m_{12}^* & m_{22}^* & m_{32}^* \\ m_{13}^* & m_{23}^* & m_{33}^* \end{pmatrix}. \qquad (3.7)$$

Look carefully at the difference between the matrix in this equation and the matrix in Eq. 3.6. You will see two differences. The most obvious is the complex conjugation of each element, but you can also see a difference in the element indices. For example, where you see $m_{23}$ in Eq. 3.6, you see $m_{32}^*$ in Eq. 3.7. In other words, the rows and columns have been interchanged.

When we change an equation from the ket form to the bra form, we must modify the matrix in two steps:

1. Interchange the rows and the columns.
2. Complex-conjugate each matrix element.

In matrix notation, interchanging rows and columns is called *transposing* and is indicated by a superscript *T*. Thus, the transpose of the matrix **M** is

$$\begin{pmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{pmatrix}^T = \begin{pmatrix} m_{11} & m_{21} & m_{31} \\ m_{12} & m_{22} & m_{32} \\ m_{13} & m_{23} & m_{33} \end{pmatrix}.$$

Notice that transposing a matrix flips it about the main diagonal (the diagonal from the upper left to the lower right).

The complex conjugate of a transposed matrix is called its *Hermitian conjugate*, denoted by a dagger. You could think of the dagger as a hybrid of the star-notation used in complex conjugation and the *T* used in transposition. In symbols,